

Transaction Control Language (TCL): COMMIT, ROLLBACK, SAVEPOINT

Detailed Hands-On Learning Guide for Corporate SQL Training

Transactions are used to manage SQL operations safely and reliably. They ensure consistency in banking systems, e-commerce applications, payment gateways, airline reservations, and enterprise systems.

Transaction Flow

```
START TRANSACTION
  ↓
SQL Operations
  ↓
COMMIT / ROLLBACK
```

Create Accounts Table

```
CREATE TABLE Accounts (
    account_id INT PRIMARY KEY,
    account_holder VARCHAR(100),
    balance DECIMAL(10,2)
);
```

Insert Sample Data

```
INSERT INTO Accounts VALUES
(101, 'Rahul Sharma', 50000),
(102, 'Anita Verma', 30000),
(103, 'Karan Singh', 45000);
```

account_id	account_holder	balance
101	Rahul Sharma	50000
102	Anita Verma	30000
103	Karan Singh	45000

1. COMMIT

COMMIT permanently saves transaction changes into the database. Once COMMIT is executed, changes cannot be undone.

```
START TRANSACTION;

UPDATE Accounts
SET balance = balance - 5000
WHERE account_id = 101;

UPDATE Accounts
SET balance = balance + 5000
WHERE account_id = 102;

COMMIT;
```

account_holder	balance
Rahul Sharma	45000
Anita Verma	35000

2. ROLLBACK

ROLLBACK cancels all changes made during the current transaction. Used when errors or failures occur.

```
START TRANSACTION;

UPDATE Accounts
SET balance = balance - 10000
WHERE account_id = 101;
```

ROLLBACK;

account_holder	balance
Rahul Sharma	50000

3. SAVEPOINT

SAVEPOINT creates checkpoints inside a transaction. Allows partial rollback instead of cancelling the complete transaction.

```
START TRANSACTION;
```

```
UPDATE Accounts  
SET balance = balance - 2000  
WHERE account_id = 101;
```

```
SAVEPOINT before_second_update;
```

```
UPDATE Accounts  
SET balance = balance + 2000  
WHERE account_id = 102;
```

```
ROLLBACK TO before_second_update;
```

```
COMMIT;
```

COMMIT vs ROLLBACK

COMMIT	ROLLBACK
Saves changes permanently	Undoes changes
Used after success	Used after failure
Data becomes final	Data restored

Hands-On Practice Exercises

- Create EmployeeSalary table
- Insert employee salary records
- Perform salary update and COMMIT
- Use ROLLBACK after incorrect update
- Use SAVEPOINT for partial rollback

Interview Questions

- What is a transaction?
- Difference between COMMIT and ROLLBACK?
- What is SAVEPOINT?
- Can ROLLBACK undo committed changes?
- Why are transactions important?

Final Summary

Command	Purpose
COMMIT	Save changes permanently
ROLLBACK	Undo changes
SAVEPOINT	Partial rollback checkpoint